

# Anticoagulation for Atrial Fibrillation with Intermediate Stroke Risk

SINGLE-AF randomized trial • Journal Club Critical Appraisal

Daehoon Kim, Young Soo Lee, Jaemin Shim, Hee Tae Yu, et al. for the SINGLE-AF Investigators

The New England Journal of Medicine • Published August 28, 2026 • DOI 10.1056/NEJMoa2607978

## TRIAL

Multicenter, open-label,  
randomized superiority trial  
18 centers • South Korea

## POPULATION

Atrial fibrillation  
CHA<sub>2</sub>DS<sub>2</sub>-VASc 1 in men / 2 in  
women

## QUESTION

DOAC therapy vs no  
anticoagulation  
24-month clinical outcomes

Presented by: DR VRUSHABH DALVI  
Department: JR1 MEDICINE

## STUDY METADATA AND SCOPE

**REGISTRATION**

ClinicalTrials.gov:  
NCT04437654

**SETTING**

18 centers in South  
Korea

**DESIGN**

Investigator-initiated  
Randomized  
superiority

**BLINDING**

Open-label  
Adjudicator-masked

**FUNDING**

Ministry of Health and Welfare, South Korea;  
Samjin Pharmaceutical; Hanmi Pharmaceutical.  
Authors report funders had no role in design,  
conduct, analysis, interpretation, or manuscript  
preparation.

**OVERSIGHT**

Independent data and safety monitoring board;  
endpoint events adjudicated by an independent  
clinical-events committee unaware of treatment  
assignment.

**Key scope issue: single-country East Asian population → external validity must be examined carefully.**

# Executive summary — the verdict upfront

Clinical question → result → absolute effect → harms → practice

## CLINICAL QUESTION

In intermediate-risk AF, does routine DOAC therapy improve net clinical outcomes versus no

## PRIMARY RESULT

0.5% vs 1.5%  
HR 0.31 (95% CI 0.10–0.94)  
P=0.03

## ABSOLUTE EFFECT

ARR  $\approx$  1.0  
percentage point  
Approx. NNT  $\approx$  100  
over 24 months

## HARM SIGNAL

Major bleeding: 0.3% vs 0.5%  
CRNM bleeding: 2.5% vs 1.6%

## CRITICAL INTERPRETATION

The primary endpoint was driven largely by fewer stroke events; cardiovascular death was 0 in both groups and major bleeding was similar. The event count was very small, making the treatment-effect estimate imprecise.

## CONSIDER IN SELECTED PATIENTS

Supports DOAC use in carefully selected intermediate-risk patients, but sparse events, open-label design, crossover, antiplatelet imbalance and single-country recruitment argue against a blanket practice change.

# Pathophysiological rationale

Why might anticoagulation help?

## ATRIAL FIBRILLATION

AF increases the risk of thromboembolism and stroke. Stroke risk varies substantially with clinical risk factors summarized by CHA<sub>2</sub>DS<sub>2</sub>-VASc.

## THE CLINICAL TENSION

At higher stroke risk, expected thromboembolic reduction generally outweighs bleeding risk. At intermediate risk, the net benefit is less certain.

## WHY THIS TRIAL?

Current U.S. and European guidance provides a Class IIa recommendation to consider oral anticoagulation in intermediate-risk patients, but randomized evidence directly comparing DOACs with no treatment was limited.

**Biological plausibility is strong; the unresolved question is the net clinical benefit at a low absolute stroke risk.**

# Historical context & evidence gap

What was known before SINGLE-AF?

## GUIDELINES

The paper states that current U.S. and European guidelines recommend considering oral anticoagulation (Class IIa) for intermediate risk: CHA<sub>2</sub>DS<sub>2</sub>-VASc 1 in men or 2 in women.

## OBSERVATIONAL DATA

Previous observational studies gave conflicting estimates of the net benefit of anticoagulation in patients with only one non-sex risk factor.

## EVIDENCE GAP

Prior phase 3 DOAC trials included some lower-risk patients but did not directly randomize intermediate-risk AF patients to DOAC vs no anticoagulation.

**Evidence gap addressed: direct randomized estimate of benefit vs no anticoagulation in intermediate-risk AF.**

# PICO framework

Translate the trial into one clinical question

## P — POPULATION

Adults  $\geq 19$  years  
Atrial fibrillation  
CHA<sub>2</sub>DS<sub>2</sub>-VASc 1 in men / 2 in women

Key exclusions: Cr  $\geq 3.5$  mg/dL or CrCl  $< 30$ ; significant liver disease; mechanical valve; moderate–severe mitral stenosis; DVT indication for anticoagulation; major structural heart disease.

## I — INTERVENTION

DOAC therapy

Apixaban 5 mg twice daily or rivaroxaban 20 mg once daily at physician discretion, with prespecified dose reductions. Other DOAC allowed for intolerance.

## C — COMPARATOR

No routine anticoagulation

Temporary DOAC permitted around cardioversion/ablation. Antiplatelet therapy permitted if clinically indicated or at physician discretion.

## O — OUTCOMES

Primary at 24 months: stroke + systemic embolism + major bleeding + cardiovascular death.

Secondary: individual components, all-cause death, TIA, CRNM bleeding, MI, hospitalization.

# Methodological architecture

Randomization, concealment, blinding and follow-up

## RANDOMIZATION

1:1 allocation  
Web-based system  
Permuted blocks of 4 or 6  
Stratified by site

## ALLOCATION

Central/web-based randomization  
supports allocation concealment.

## BLINDING

Open-label treatment  
Independent, blinded endpoint adjudication

## FOLLOW-UP

Baseline, 3, 6, 12, 18 and 24 months  
Median follow-up: 730 days  
94.6% completed 24 months in each arm

**Major validity trade-off: endpoint adjudication was masked, but treatment decisions and patient/clinician behavior were not.**

## STRENGTH

Randomization + site stratification + blinded adjudication reduce major sources of bias.

## VULNERABILITY

Open-label design leaves room for differential co-interventions, adherence, symptom reporting and clinical management.

# Endpoint hierarchy & composite audit

The primary endpoint is statistically positive — but what is inside it?

## Stroke

3 vs 10  
HR 0.30 (0.08–1.08)

## Systemic embolism

0 vs 1  
Very sparse

## Major bleeding

3 vs 4  
HR 0.75 (0.17–3.35)

## CV death

0 vs 0  
No events

**Composite result: 4/902 (0.5%) vs 13/901 (1.5%); HR 0.31 (95% CI 0.10–0.94); P=0.03.**

## AUDIT QUESTION

Are components equally patient-important?

## APPRAISAL

No: the apparent benefit is primarily thromboembolic; there were no cardiovascular deaths and major bleeding was similar.



# Statistical plan & sample size

Was the trial powered for what it ultimately observed?

## ASSUMED EVENT RATES

DOAC: 2.0%  
No anticoagulation:  
4.5%  
Target difference: 2.5  
percentage points

## PLANNED SAMPLE

1,800 patients  
900 per group  
80% power  
Two-sided  $\alpha=0.05$   
12% attrition assumed

## ANALYSIS

Intention-to-treat  
Aalen-Johansen  
cumulative  
incidence  
Fine-Gray  
subdistribution  
hazard model

## MULTIPLICITY

No adjustment for  
secondary or  
subgroup analyses.  
Their CIs should not  
be used as definitive  
treatment-effect  
evidence.

**Observed primary events: only 17 total (4 vs 13) — far fewer than the planning assumptions imply.**

## KEY STATISTICAL CONCERN

The trial met its nominal significance threshold, but the unexpectedly low event rate produces a wide treatment-effect CI and makes the magnitude of benefit fragile.

# Competing risks & analysis choices

Why Aalen-Johansen and Fine-Gray were used

## COMPETING EVENT

The primary endpoint includes cardiovascular death. Noncardiovascular death can prevent a patient from experiencing the primary event and was treated as a competing event.

## CUMULATIVE INCIDENCE

The Aalen-Johansen method estimates cumulative incidence in the presence of competing risks.

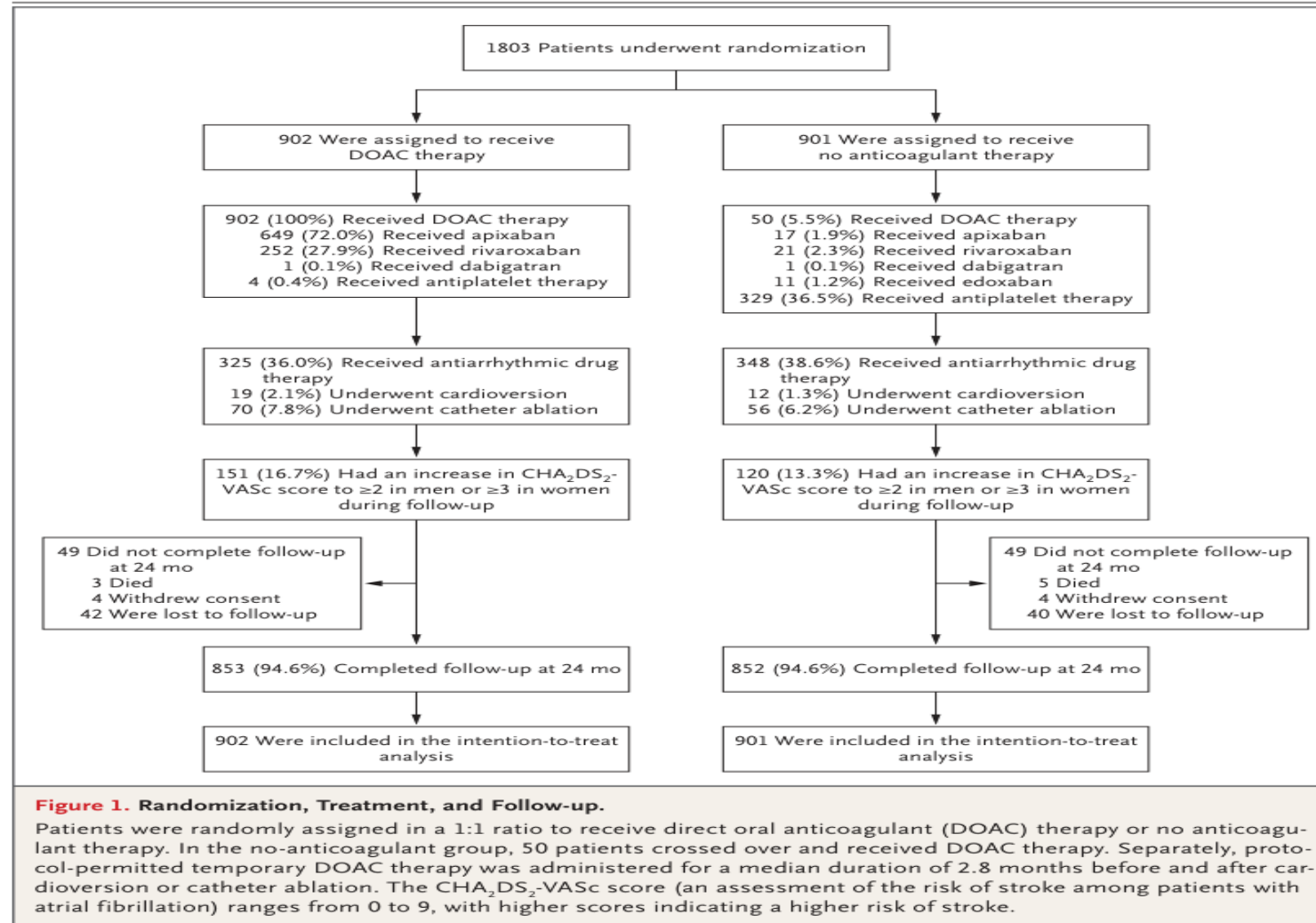
## TREATMENT EFFECT

Fine-Gray subdistribution hazard models estimate the relative effect on cumulative incidence while accounting for the competing event.

The paper reports no evidence of violation of proportional-hazards assumptions ( $P=0.18$  for the Cox model sensitivity assessment).

# Participant flow — CONSORT audit

Randomization, crossover, follow-up and analysis population  
myocardial infarction or deaths from cardiovascular disease, 0.10; 95% CI, 0.01 to 0.77) (Fig. S3).



**1,803  
randomized**

**49 + 49  
did not  
complete  
follow-up**

**94.6%  
completed  
follow-up  
in each  
arm**

**50 (5.5%)  
crossed  
over from  
no-  
anticoagulant  
to  
DOAC**

# Participant flow — what matters for validity?

The headline is reassuring, but look at treatment exposure

## FOLLOW-UP

94.6% completed 24 months in both arms.  
5.4% incomplete in each group.

## CROSSOVER

50/901 (5.5%) in the no-anticoagulant group received a DOAC.

## CO-INTERVENTION

Antiplatelet use:  
36.5% vs 0.4%.

## ITT

All randomized patients remained in the ITT analysis.

**Interpretation: crossover tends to dilute a true treatment effect, while differential antiplatelet use complicates bleeding comparisons.**

## IMPORTANT NUANCE

The no-anticoagulant group was not pharmacologically “untreated”: one-third received antiplatelet therapy, and temporary DOAC around rhythm-control procedures was allowed.

# Baseline characteristics — source table

High-level balance appears good: clinically meaningful differences are modest

| Characteristic                                             | DOAC Therapy (N=902) | No Anticoagulant Therapy (N=901) |
|------------------------------------------------------------|----------------------|----------------------------------|
| <b>Age</b>                                                 |                      |                                  |
| Mean — yr                                                  | 60.7±7.8             | 60.0±8.1                         |
| ≥65 yr — no. (%)                                           | 304 (33.7)           | 282 (31.3)                       |
| Male sex — no. (%)                                         | 682 (75.6)           | 694 (77.0)                       |
| <b>Type of atrial fibrillation — no. (%)</b>               |                      |                                  |
| Paroxysmal                                                 | 636 (70.5)           | 658 (73.0)                       |
| Nonparoxysmal                                              | 266 (29.5)           | 243 (27.0)                       |
| Median time since atrial fibrillation diagnosis (IQR) — mo | 21 (2–60)            | 21 (2–58)                        |
| <b>Medical history — no. (%)</b>                           |                      |                                  |
| Hypertension                                               | 435 (48.2)           | 482 (53.5)                       |
| Diabetes mellitus                                          | 67 (7.4)             | 70 (7.8)                         |
| Heart failure                                              | 93 (10.3)            | 72 (8.0)                         |
| Stroke or transient ischemic attack                        | 0 (0)                | 2 (0.2)                          |
| Dyslipidemia                                               | 224 (24.8)           | 193 (21.4)                       |
| Myocardial infarction                                      | 3 (0.3)              | 2 (0.2)                          |
| Peripheral artery disease                                  | 8 (0.9)              | 4 (0.4)                          |
| Previous ablation for atrial fibrillation — no. (%)        | 266 (29.5)           | 285 (31.6)                       |
| Previous cardioversion — no. (%)                           | 162 (18.0)           | 137 (15.2)                       |
| Current alcohol use — no. (%)                              | 253 (28.0)           | 265 (29.4)                       |
| Current smoker — no. (%)                                   | 97 (10.8)            | 123 (13.7)                       |
| <b>CHA<sub>2</sub>DS<sub>2</sub>-VASc score†</b>           |                      |                                  |
| Mean                                                       | 1.3±0.5              | 1.3±0.5                          |
| Median (IQR)                                               | 1 (1–1)              | 1 (1–1)                          |
| <b>CHA<sub>2</sub>DS<sub>2</sub>-VA score‡</b>             |                      |                                  |
| Mean                                                       | 1.0±0.1              | 1.0±0.2                          |
| Median (IQR)                                               | 1 (1–1)              | 1 (1–1)                          |
| <b>CHADS<sub>2</sub> score§</b>                            |                      |                                  |
| Mean                                                       | 0.7±0.5              | 0.7±0.5                          |
| Median (IQR)                                               | 1 (0–1)              | 1 (0–1)                          |
| <b>HAS-BLED score¶</b>                                     |                      |                                  |
| Mean                                                       | 0.5±0.6              | 0.5±0.7                          |
| Median (IQR)                                               | 0 (0–1)              | 0 (0–1)                          |
| <b>Blood pressure — mm Hg</b>                              |                      |                                  |
| Systolic                                                   | 125.6±15.4           | 127.1±14.4                       |
| Diastolic                                                  | 76.6±12.4            | 77.2±11.8                        |

## AGE

60.7 vs 60.0 years; ≥65 years 33.7% vs 31.3%

## AF PHENOTYPE

Paroxysmal AF 70.5% vs 73.0%; nonparoxysmal 29.5% vs 27.0%

## RISK SCORES

CHA<sub>2</sub>DS<sub>2</sub>-VASc mean 1.3 in both; HAS-BLED mean 0.5 in both

## CLINICAL NOTE

Heart failure 10.3% vs 8.0%; hypertension 48.2% vs 53.5%.

# Baseline characteristics — appraisal

Do not use baseline P-values as a balance test

## WHAT IS REASSURING

Groups were broadly similar in age, sex, AF type, CHA<sub>2</sub>DS<sub>2</sub>-VASc, HAS-BLED, blood pressure and renal function.

## WHAT DESERVES ATTENTION

Hypertension was 48.2% vs 53.5%; heart failure 10.3% vs 8.0%; prior cardioversion and ablation differed modestly. None is obviously large enough to overturn randomization.

## METHOD PRINCIPLE

Randomized trials do not require hypothesis tests for baseline balance. Ask whether any imbalance is clinically important and strongly prognostic.

**Overall: baseline imbalance is not the major threat to internal validity here.**

# Treatment exposure & co-interventions

The intervention was not a single DOAC regimen

## DOAC ARM

902/902 received DOAC therapy.  
Apixaban 72.0%  
Rivaroxaban 27.9%  
Dabigatran 0.1%

## DOSE INTENSITY

81.6% received full doses; 18.4% received reduced doses.

## CONTROL ARM

50/901 (5.5%) received DOAC therapy.

## ANTIPLATELET

0.4% in DOAC arm vs 36.5% in no-anticoagulant arm.

**This is a pragmatic comparison of “DOAC strategy” vs “no routine anticoagulation strategy,” not a head-to-head drug trial.**

## APPRAISAL

Heterogeneous DOAC exposure improves real-world relevance but makes the result less attributable to one specific molecule or dose.

# Primary & secondary outcomes — source table

Read the absolute numbers before the hazard ratio

**Table 2.** Primary, Secondary, and Exploratory End Points at 24 Months.\*

| End Point                                                                                   | DOAC Therapy<br>(N=902)                          | No Anticoagulant Therapy<br>(N=901) | Difference<br>(95% CI)           | Hazard Ratio<br>(95% CI) |
|---------------------------------------------------------------------------------------------|--------------------------------------------------|-------------------------------------|----------------------------------|--------------------------|
|                                                                                             | <i>no. of patients (estimated %)<sup>†</sup></i> |                                     | <i>percentage points</i>         |                          |
| <b>Primary end point</b>                                                                    |                                                  |                                     |                                  |                          |
| Composite of stroke, systemic embolism, major bleeding, or death from cardiovascular causes | 4 (0.5)                                          | 13 (1.5)                            | −1.0 (−2.0 to −0.1) <sup>‡</sup> | 0.31 (0.10 to 0.94)      |
| <b>Secondary end points<sup>§</sup></b>                                                     |                                                  |                                     |                                  |                          |
| Stroke                                                                                      | 3 (0.3)                                          | 10 (1.1)                            | −0.8 (−1.6 to 0.0)               | 0.30 (0.08 to 1.08)      |
| Ischemic event                                                                              | 1 (0.1)                                          | 10 (1.1)                            | −1.0 (−1.8 to −0.3)              | 0.10 (0.01 to 0.77)      |
| Hemorrhagic event                                                                           | 2 (0.2)                                          | 0                                   | 0.2 (−0.1 to 0.6)                | —                        |
| Systemic embolism                                                                           | 0                                                | 1 (0.1)                             | −0.1 (−0.3 to 0.1)               | —                        |
| Major bleeding <sup>¶</sup>                                                                 | 3 (0.3)                                          | 4 (0.5)                             | −0.1 (−0.7 to 0.5)               | 0.75 (0.17 to 3.35)      |
| Intracranial hemorrhage <sup>¶</sup>                                                        | 2 (0.2)                                          | 1 (0.1)                             | 0.1 (−0.3 to 0.5)                | 2.00 (0.18 to 22.05)     |
| Gastrointestinal hemorrhage                                                                 | 1 (0.1)                                          | 2 (0.2)                             | −0.1 (−0.5 to 0.3)               | 0.50 (0.05 to 5.51)      |
| Fatal bleeding                                                                              | 0                                                | 0                                   | —                                | —                        |
| Death from cardiovascular causes                                                            | 0                                                | 0                                   | —                                | —                        |
| Clinically relevant nonmajor bleeding <sup>¶</sup>                                          | 22 (2.5)                                         | 14 (1.6)                            | 0.9 (−0.4 to 2.2)                | 1.58 (0.81 to 3.08)      |
| Death from any cause                                                                        | 3 (0.3)                                          | 5 (0.6)                             | −0.2 (−0.9 to 0.4)               | 0.60 (0.14 to 2.51)      |
| Transient ischemic attack                                                                   | 0                                                | 3 (0.3)                             | −0.3 (−0.7 to 0.0)               | —                        |
| Myocardial infarction                                                                       | 0                                                | 0                                   | —                                | —                        |
| Hospitalization for any cause                                                               | 163 (18.6)                                       | 154 (17.6)                          | 0.9 (−2.7 to 4.6)                | 1.07 (0.86 to 1.33)      |
| <b>Exploratory end points<sup>§</sup></b>                                                   |                                                  |                                     |                                  |                          |
| Composite of ischemic stroke or systemic embolism**                                         | 1 (0.1)                                          | 10 (1.1)                            | −1.0 (−1.8 to −0.3)              | 0.10 (0.01 to 0.77)      |
| Composite of major bleeding or clinically relevant nonmajor bleeding <sup>¶</sup>           | 24 (2.7)                                         | 17 (2.0)                            | 0.8 (−0.6 to 2.2)                | 1.42 (0.76 to 2.63)      |

\* The analyses were performed in the intention-to-treat population, which included all the patients who underwent randomization, grouped according to their assigned trial group; the start time of follow-up for the intention-to-treat analysis was the day of randomization. For end points that included death from cardiovascular causes as a component, including the primary end point, death from noncardiovascular causes was treated as a competing event. For other end points that do not include death from any cause, death from any cause was treated as a competing event. The hazard ratios were calculated with the use of subdistribution-hazard models from a competing risk analysis.

<sup>†</sup> The 24-month cumulative incidence of end-point events subject to competing risks was estimated with the cumulative-incidence function, whereas the cumulative incidence of death from any cause was estimated with the Kaplan–Meier method. Therefore, the reported percentages may not correspond directly to the ratio of the numerator to the denominator.

<sup>‡</sup> P=0.03 for superiority of DOAC therapy to no anticoagulant therapy, calculated with Gray's test.

<sup>§</sup> Analyses of the exploratory end points were performed post hoc. The widths of the 95% confidence intervals have not been adjusted for multiple comparisons and should therefore not be used to infer treatment effects.

<sup>¶</sup> Bleeding events were defined in accordance with the criteria of the International Society on Thrombosis and Haemostasis.

<sup>¶</sup> One patient in the no-anticoagulant group had a subdural hemorrhage, which was classified as intracranial hemorrhage but not as hemorrhagic stroke.

\*\* One patient in the no-anticoagulant group had both an ischemic stroke and systemic embolism.

## PRIMARY

0.5% vs 1.5%  
ARR ≈ 1.0 pp

## STROKE

0.3% vs 1.1%  
HR 0.30; CI crosses 1

## CRNM BLEEDING

2.5% vs 1.6%  
HR 1.58; CI crosses 1

## HOSPITALIZATION

18.6% vs 17.6%  
No clear benefit



# Effect size: statistical vs clinical significance

The absolute effect is small because baseline event risk is small

## RELATIVE EFFECT

HR 0.31  
95% CI 0.10–0.94  
≈69% relative  
reduction in hazard

## ABSOLUTE EFFECT

1.0 percentage-point  
difference at 24  
months

## NNT

Approx. 100 patients  
treated for 24  
months to prevent  
one additional  
primary-endpoint  
event, using the  
absolute cumulative-  
incidence difference.

## PRECISION

95% CI for absolute  
difference: –2.0 to –0.1  
pp.  
The CI is close to the null  
and the total event count  
is only 17.

**Clinical significance question: is preventing roughly 1 composite event per 100 treated over 2 years worth routine anticoagulation for this patient?**

## ANSWER

Potentially — particularly if stroke prevention is highly valued — but the decision remains preference-sensitive because absolute stroke risk is low and minor clinically relevant bleeding was numerically higher.

# Deconstructing the benefit

What actually changed?

## STROKE

3 vs 10 events  
0.3% vs 1.1%  
HR 0.30 (0.08–1.08)  
Statistical uncertainty remains.

## SYSTEMIC EMBOLISM

0 vs 1 event  
Too few events for stable inference.

## MAJOR BLEEDING

3 vs 4 events  
0.3% vs 0.5%  
No clear excess or reduction.

**Cardiovascular death: 0 in both groups. Myocardial infarction: 0 in both groups.**

## BOTTOM LINE

The positive composite does not demonstrate a mortality benefit. It mainly reflects fewer thromboembolic events, especially stroke, in a setting where the absolute event rate was very low.

# Cumulative incidence of the primary endpoint

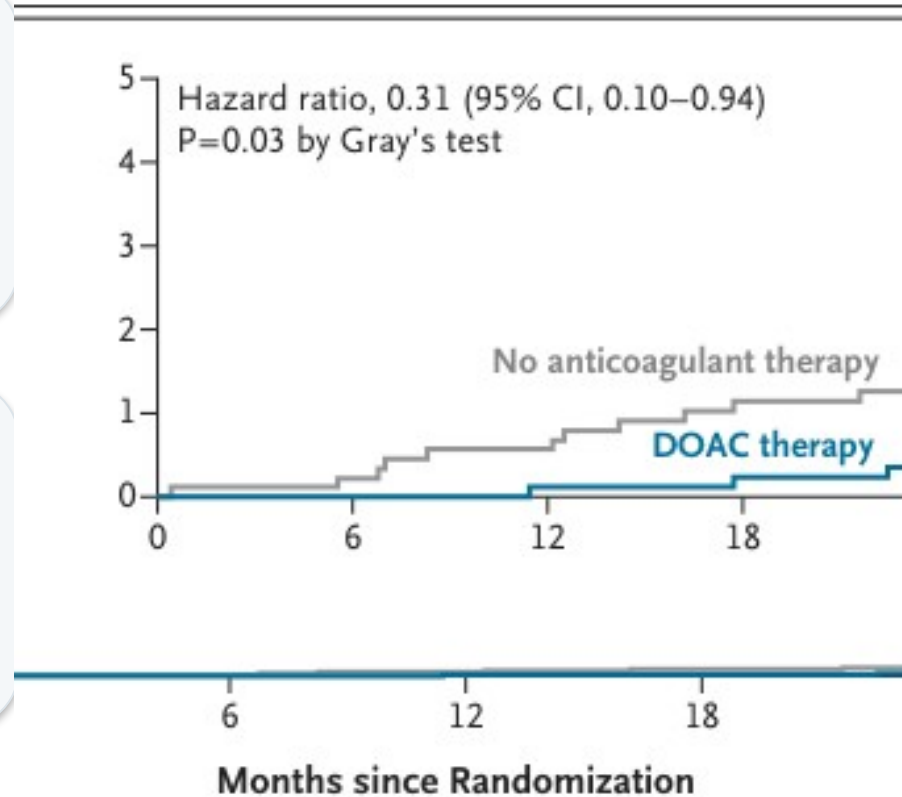
Use the curve — but read the y-axis and number-at-risk table

## CURVE

Curves separate gradually, with the no-anticoagulant group accumulating more events.

## NUMBER AT RISK

At 24 months: 840 no-anticoagulant and 849 DOAC patients remained at risk in the displayed table.



## STATISTIC

Subhazard ratio 0.31  
95% CI 0.10–0.94  
Gray's test P=0.03

## CAUTION

Only 17 primary events occurred. A visually separated curve can still represent an imprecisely estimated effect.

|                          |     |     |     |     |
|--------------------------|-----|-----|-----|-----|
|                          | 6   | 12  | 18  | 24  |
| No anticoagulant therapy | 877 | 856 | 843 | 840 |
| DOAC therapy             | 885 | 859 | 853 | 849 |

# Subgroup analysis — source forest plot

The NEW ENGLAND JOURNAL of MEDICINE

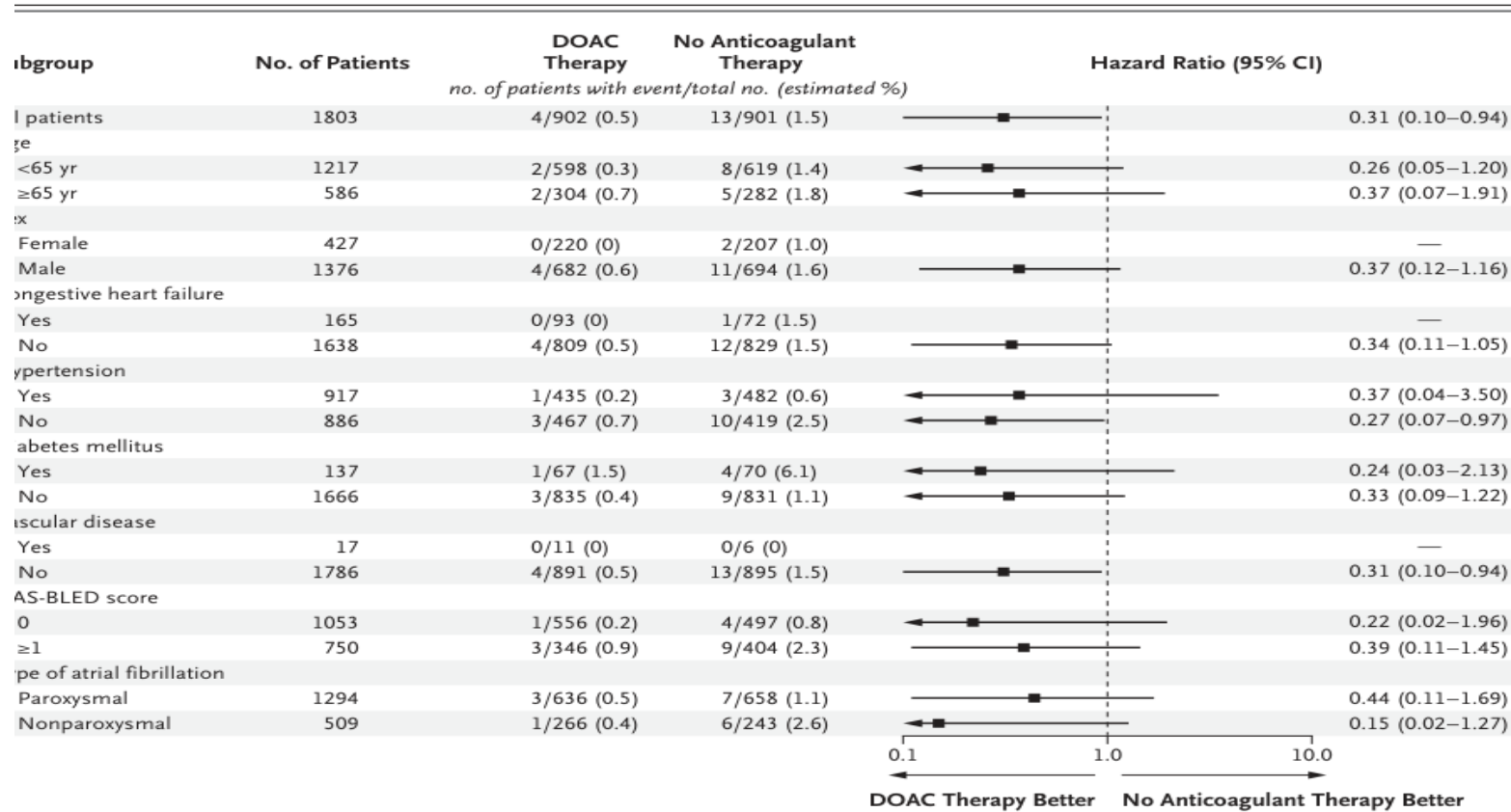


Figure 3. Subgroup Analyses of the Primary End Point.

Shown are hazard ratios for the comparison between DOAC therapy and no anticoagulant therapy for a primary end-point event according to prespecified subgroups. Death from noncardiovascular causes was treated as a competing event. Hazard ratios were estimated with the use of subdistribution-hazard models, and the cumulative incidence was estimated with the cumulative-incidence function. The HAS-BLED score is a measure of the risk of bleeding among patients with atrial fibrillation who are receiving anticoagulant therapy (scores range from 0 to 9, with higher scores indicating a greater risk). Because the statistical analysis plan did not include a provision

nate

## OVERALL

HR 0.31 (0.10–0.94)

## POST HOC

AF type (paroxysmal vs nonparoxysmal) analysis was post hoc.

## MULTIPLICITY

No multiplicity correction; subgroup CIs should not establish definitive differences.

# Subgroup appraisal

What can — and cannot — be concluded?

## WHAT THE FIGURE SHOWS

Point estimates generally favor DOAC therapy across displayed subgroups, but many confidence intervals are wide and cross 1.

## WHAT IT DOES NOT SHOW

The forest plot alone does not prove that age, sex, hypertension, diabetes, HAS-BLED or AF type modifies treatment effect.

## METHOD RULE

A credible subgroup claim requires a statistically persuasive treatment-by-subgroup interaction, not simply  $P < 0.05$  within one subgroup. The paper itself warns that multiplicity was not adjusted.

**Subgroup conclusion: hypothesis-generating, not practice-defining.**

# Safety, harms & adverse events

Benefit must be weighed against bleeding and treatment burden

## SERIOUS ADVERSE EVENTS

8.9% vs 9.3%  
80 vs 84 patients

## MAJOR BLEEDING

0.3% vs 0.5%  
3 vs 4 patients  
HR 0.75 (0.17–3.35)

## CRNM BLEEDING

2.5% vs 1.6%  
22 vs 14 patients  
HR 1.58 (0.81–3.08)

## ICH

2 vs 1 patients  
Very sparse; HR 2.00  
(0.18–22.05)

**Approx. NNH for CRNM bleeding  $\approx$  111 over 24 months, but the CI includes no difference.**

## IMPORTANT

The no-anticoagulant arm used antiplatelet therapy in 36.5% vs 0.4% in the DOAC arm, potentially disadvantaging the control group for bleeding comparisons.

# Internal validity — strengths & vulnerabilities

Can I trust the result?

## STRENGTHS

- Randomized 1:1 allocation
- Site-stratified web randomization
- ITT primary analysis
- Independent blinded endpoint adjudication
- Prespecified primary analysis
- Similar follow-up completion
- Proportional-hazards assumptions assessed

## VULNERABILITIES

- Open-label design
- 50 control patients received DOAC
- Differential antiplatelet use
- 5.4% did not complete 24-month follow-up
- Very low event rate
- Primary endpoint amended in Jan 2025
- Multiple secondary/subgroup analyses without multiplicity correction

## OVERALL JUDGMENT

Internal validity: reasonably strong for the primary randomized comparison, but confidence in the magnitude of benefit is reduced by sparse events and design/management asymmetries.

**Verdict: trustworthy direction of effect; less certain magnitude of effect.**

# Cochrane Risk of Bias 2 — structured appraisal

A practical RoB 2 judgment for this trial

|                                              |                      |                                                                                                         |
|----------------------------------------------|----------------------|---------------------------------------------------------------------------------------------------------|
| <b>Randomization process</b>                 | <b>Low risk</b>      | Central web randomization; balanced baseline profile.                                                   |
| <b>Deviations from intended intervention</b> | <b>Some concerns</b> | Open-label; 5.5% control crossover; differential co-interventions.                                      |
| <b>Missing outcome data</b>                  | <b>Some concerns</b> | 5.4% in each arm did not complete 24 months.                                                            |
| <b>Measurement of outcome</b>                | <b>Low risk</b>      | Independent adjudication masked to treatment assignment; objective endpoints.                           |
| <b>Selection of reported result</b>          | <b>Some concerns</b> | Primary endpoint amended in 2025; multiple secondary/subgroup analyses without multiplicity correction. |

**Overall RoB 2 judgment: SOME CONCERNS — chiefly from open-label management, crossover, incomplete follow-up, and reporting/endpoint considerations.**



# GRADE certainty of evidence

Suggested certainty for the clinically important primary outcome

## STARTING POINT

Randomized trial →  
HIGH certainty

## RISK OF BIAS

Downgrade 1 level  
Open-label +  
crossover +  
incomplete follow-up

## IMPRECISION

Downgrade 1 level  
Only 17 primary  
events; wide effect  
CI

## INDIRECTNESS

Potential concern for Indian  
practice because all  
participants were East Asian  
and recruited in one  
country.

# Guideline integration

Where does SINGLE-AF fit?

## BEFORE THE TRIAL

The paper states that U.S. and European guidelines gave a Class IIa recommendation to consider oral anticoagulation in intermediate-risk AF (CHA<sub>2</sub>DS<sub>2</sub>-VASc 1 in men; 2 in women).

## WHAT SINGLE-AF ADDS

Direct randomized evidence supporting a lower composite event rate with DOAC therapy vs no anticoagulation over 24 months.

**Does it overturn the guideline position? Not clearly. It strengthens the evidence base behind “consider,” rather than proving universal anticoagulation.**

# Trial chronology & endpoint amendment

A detail worth asking the presenters about

**2020**

Enrollment began  
July 27, 2020

**JAN 2025**

Independent DSMB  
noted deaths were  
noncardiovascular  
and unlikely  
treatment-related.

**AMENDMENT**

Primary endpoint  
changed from all-  
cause death to  
cardiovascular  
death; all-cause  
death became  
secondary.

**2026**

Publication  
August 28, 2026  
NEJM

**Why this matters: the amendment occurred before final analysis and was made while the DSMB was unaware of treatment assignment, but it still changes the endpoint hierarchy.**

# Local applicability — Indian practice

Can I apply this to a North Indian hospital tomorrow?

## PATIENT MATCH

Trial patients: mean age ~60; 23.7% women; 71.8% paroxysmal AF; low HAS-BLED; preserved renal function on average.

Older, frailer, CKD, higher bleeding-risk patients may differ substantially.

## ETHNIC / GEOGRAPHIC

All participants were East Asian and from South Korea. Stroke and bleeding risk profiles can differ across populations.

## FEASIBILITY

DOAC availability is generally practical, but cost, adherence, renal monitoring, dose selection and follow-up remain real-world considerations.

## PRACTICALITY

Decision should integrate stroke risk, bleeding risk, patient preference, drug affordability and anticipated follow-up.

**External validity: MODERATE at best — the biological question is generalizable; the absolute risk estimate may not be.**

# Actionable takeaways by care setting

What changes in practice?

## EMERGENCY / WARD

Confirm AF and calculate stroke risk correctly.

Do not treat the trial as a mandate for every intermediate-risk patient.

Assess bleeding risk, renal/liver function, concomitant antiplatelets and upcoming procedures.

## ICU / HIGH DEPENDENCY

This trial is not an ICU trial. Avoid extrapolating its findings to critically ill, unstable or substantially higher-risk populations.

## OPD / SPECIALIST

For an intermediate-risk patient, DOAC therapy is reasonable to consider, especially when stroke prevention is prioritized and bleeding risk is low. Shared decision-making remains central.

**Practice impact: reinforces “consider” rather than “automatically anticoagulate.”**

## Question ?

### Clinical

Would you prescribe a DOAC to a 60-year-old man with CHA<sub>2</sub>DS<sub>2</sub>-VASc 1 after seeing this trial?

# What would change my mind?

Predefine the evidence needed to move from “consider” to “practice-changing”

## MORE EVENTS

A larger trial with enough stroke and bleeding events to narrow the confidence intervals.

## REPLICATION

Independent multicenter replication across regions and ethnic groups.

## HARD OUTCOMES

Consistent reduction in ischemic stroke with reassuring major-bleeding and mortality data.

## LOCAL DATA

Indian/South Asian evidence with realistic affordability and adherence patterns.

**The key is not a smaller P-value — it is a more precise, clinically important and generalizable estimate of net benefit.**

## TARGET

A future study should preserve rigorous randomization while minimizing crossover and clearly separating thromboembolic benefit from bleeding harm.

# Conclusion — the four questions

Journal Club should end with a verdict, not a summary

## 1. CAN I TRUST IT?

Yes, with reservations. Randomization and blinded adjudication are strengths; sparse events and open-label management reduce certainty.

## 2. DOES IT MATTER?

Potentially. ARR  $\approx$  1.0 pp, but no mortality benefit and very low absolute event rates.

## 3. CAN I APPLY IT?

Only selectively. Single-country East Asian recruitment limits direct generalization to Indian practice.

## 4. WILL IT CHANGE TOMORROW?

It strengthens the case to consider a DOAC in intermediate-risk AF; it does not justify universal treatment based on this trial alone.

## FINAL PRACTICE VERDICT

**CONSIDER IN SELECTED PATIENTS**

**SINGLE-AF supports DOAC therapy in intermediate-risk AF, but confirmatory evidence is needed to define the true magnitude of net clinical benefit.**